



**PERTANDINGAN KEMAHIRAN KOLEJ VOKASIONAL  
PERINGKAT KEBANGSAAN 2024**

## **BIDANG\_IT\_NETWORK SYSTEMS ADMINISTRATION**

**SOALAN PRAKTIKAL**

**MASA : 6 JAM**

**NAMA:** \_\_\_\_\_

**NO K/P:** \_\_\_\_\_

**INSTITUSI:** \_\_\_\_\_

## Introduction

The competition has a fixed start and finish time. You must decide how to best divide your time.

Please **carefully read** the following instructions!

When the competition time ends, please leave your station in a running state. The assessment will be done in the state as it is. No reboot will be initiated as well as powered off machines will not be powered on! The machines involved in the automation except MGMT will be reverted to initial state for marking purposes. The default configuration for Cisco devices is included.

Set all user password and credential with Skills39 unless being specifically stated with password.

Please use the information below for all the servers and clients.

This Test Project Proposal consists of the following documentation/files:

- ITNSA-KVSkills2024.pdf
- debian-12.5.0-amd64-DLBD-1.iso
- hosts
- ansible.cfg
- AD-Users.csv

### LOGIN

|                   |                       |
|-------------------|-----------------------|
| Username Linux:   | root / itnsa          |
| Username Windows: | Administrator / itnsa |
| Username Cisco:   | itnsa                 |
| Password:         | Skills39              |

All VMs and devices are connected to the management network (10.1.1.0/24) and have a statically configured IP address. The management network will be used for configuring the different hosts. You can login using username and password over SSH or WINRM. You may install any additionally required packages and features on the VMs. The ISO for Debian is included.

### System Configuration

|                  |                     |
|------------------|---------------------|
| Region/timezone: | Asia / Kuala Lumpur |
| Locale:          | English US (UTF-8)  |
| Key Map:         | English US          |

When tasked with configuring SSL or TLS, you use certificates server in the topology. If unable to do so, you can use a self-signed-certificate but you will lose some mark. Please take care to hide any warnings.

## Description of project and tasks

You are the IT consultant responsible for KVSILLS2024 and you have been tasked with implementing a complex IT-environment.

## PART A: WINDOWS SERVER

### Work Task DC Server

**NOTE: Please use the default configuration if you are not given the details.**

- This server is already preinstalled (Windows Server 2019 with GUI)
- Configure the server with the settings specified in the diagram at the end of the document
- Modify the default Firewall rules to allow ICMP (ping) traffic

### *Active Directory Domain Service (ADDS)*

- Install and Configure Active Directory Domain Services
  - Set this server as the primary domain controller and Global catalogue for **kvskills.my**
- Create the following Organizational Units:
  - Managers
  - Competitors
  - Visitors
- Create the following Global AD Groups:
  - KV-Managers
  - KV-Competitors
  - KV-Visitors
- Import users from AD-Users.csv file located in C:\ on DC server.
  - Account placed in appropriate OU
  - Accounts is enabled with all properties in CSV file
  - User principal name with @kvskills.my suffix
  - User is not required to change password at first login.
  - All users have to use "\\CORE\homes\%username%" as their home drive. Use G:\ as drive letter.

### *Domain Name System (DNS)*

- Install and configure DNS Service
  - Create static A records for all servers of kvskills.my domain
  - Create PTR reverse zone for the internal subnet
  - Create CNAME records required by services
  - Configure conditional forwarder with itnsa.my
  - Configure records for NCSI server for internet simulation (msftncsi.com, msftconnecttest.com)

### *Active Directory Certificate Services (ADCS)*

- Install and configure Certificate Service
  - Install only "Certificate Authority"
  - Create enterprise root certificate with the following properties
    - Common Name = KVSkills-CA
  - Create necessary templates for Web Server and User usage.

## GPO

- Install and Configure Policy Management
- Setup the following settings:
- All users should receive a login banner that reads
  - Title: "Welcome to KVSkills 2024"
  - Message: "Only authorized personnel allowed to access"
- Disable all first sign-in animation for first time login users.
- Map the shared folder of H:\shares from CORE as shared drive with the letter I:
- Autoenrollment of the "KV-ClientServerCert" Certificate to all client users.

## Work Task CLIENT

**NOTE: Please use the default configuration if you are not given the details.**

- This client is already preinstalled (Windows 10 Enterprise Edition)
- Configure the client with the settings specified in the diagram at the end of the document
- Modify the default Firewall rules to allow ICMP (ping) traffic
- Enable and set the local administrator password to **Skills39**
- Join the computer to the **kvskills.my** domain
- Set the power configuration so the client will never go to sleep while plugged in
- Install outlook/thunderbird and configure mailbox for user kv10
- Send/Reply email to user kv20

## PART B: LINUX SERVER

### Work Task LNX1 Server

**NOTE:** Please use the default configuration if you are not given the details.

- The base Debian OS has been set up on LINUX SERVER
- Configure the server with the hostname, domain and IP specified in the appendix

### Domain Name System

- Install and configure bind9
- Configure itnsa.my name server
- Create static A and PTR records for all servers of itnsa.my
- Create CNAME records required by services
- Configure conditional forwarder with kvskills.my to windows server
- Set the linux server to use own address as DNS server

### MAIL

- Install and configure e-mail server (postfix , dovecot)
- Create user kv10 to kv20 with password "Skills39"
- Make sure kv10 to kv20 have access via IMAPS and SMTPS
- Use certificate signed by configured ADCS for SSL/TLS encryption

## PART C: NETWORK

### SWITCH

- This part will involve **SW1 and SW2** only. SW3 is configured using automation
- Configure SVI interface based on information in IP addressing table
- Create trunking port between switches.
  - Set appropriate access port for client endpoints
- Configure LACP for link between SW1, SW2 and SW3
  - All links should be in active mode.
- Configure the Virtual Trunking Protocol (VTP)
  - Set the SW2 as VTP server. The others switch will be the VTP Client.

|              |             |
|--------------|-------------|
| VTP Domain   | kvskills.my |
| VTP Password | Skills39    |
| VTP Version  | 3           |

|        |            |
|--------|------------|
| VLAN10 | NAME: SRV  |
| VLAN20 | NAME: LAN  |
| VLAN99 | NAME: MGMT |

## PART D: AUTOMATION USING ANSIBLE (EVE-NG)

### Basic Configuration

1. Install and setup Ansible on the involved hosts in the topology. There is a preconfigured hosts file given that contains the information of every host. DO NOT CHANGE THIS FILE.
2. Create a folder at /data/ansible for the tasks configuration. All playbooks should be on the root of the directory /data/ansible on MGMT VM. Feel free to add or create any file/folder for running the playbook.
3. For marking, all playbooks will be run in order from each part's respective directory using the command "ansible-playbook *playbookname.yml*".
4. All tasks should have state of "ok" or "skipped" even after running for more than one time.
5. "All hosts" will be referring to the devices of LNX2, CORE, HQ, ISP, SW3

### 00- facts.yml

- Create a playbook called 00-facts.yml to test the connectivity of devices on all hosts.
  - Configure to test to all connection to all hosts (ping, ios\_facts, win\_ping)

### 01- hostname.yml

- Create a playbook called 01-hostname.yml for configuring hostname and domain name
  - All hosts should received the hostname
    - It should be based on the network table below.
    - An variable of hostname is configured in ansible for convenience.
  - Linux hosts should have the domain name of itnsa.my
  - Network devices should have the domain name of KVSkills.my

### 02- ipaddress.yml

- Create a playbook called 02-ipaddress.yml for configuring IP address.
  - Configure IP address based on the IP address table including the sub-interfaces.
  - Configure primary DNS server address.
  - **DO NOT** modify the Management network interfaces.

### 03- switch.yml

- Create a playbook called 03-switch.yml for configuring switching
  - Configure LACP for link between SW1 and SW2 from SW3
    - SW3 will be in passive mode for both link
  - Create trunking ports between switches.
  - Configure to enable VTP client

### 04- ospf.yml

- Create a playbook called 04-ospf.yml for configuring routing using OSPF.
  - Configure OSPF for the network, use OSPF 100 and area 0 by default
    - Make sure all virtual machines in network can communicate with each other.
    - **DO NOT** advertise management network
    - Protect OSPF link with md5 authentication using password "Skills39"

## 05- net-dhcp.yml

- Create a playbook called 05-net-dhcp.yml for configuring DHCP on router.
  - Create DHCP scope with the following parameter for client subnet
    - The first IP should be 192.168.10.120/24
  - Set appropriate value for scope options of DNS and Gateway

## 06- adds.yml

- Create a playbook called 06-adds.yml for installation of Active Directory Domain Services
  - Configure CORE as the secondary domain controller (No Global Catalogue)

## 07- files.yml

- Create a playbook called 07-files.yml for configuring file sharing on Windows
  - Configure the RAID 5 for the disk.
    - The disk and formatting can be done manually.
    - Make sure to create a snapshot after disk is created before proceeding.
    - The three 1G disk is should be installed on the VM.
    - Format the disks as NTFS and mount the drive as H:\.
- Configure the file share of home folder.
  - Create a directory with the local path of H:\homes and share folder
  - Create a directory with the local path of H:\shares and make it publicly accessible

## 08- Inxweb.yml

- Create a playbook called 08-Inxweb.yml for configuring the Apache Web Server services.
  - Make the webpage is secured with TLS/SSL certificated generated from ADCS
    - The website on LNX2 with URL of https://www.itnsa.my showing the following.
    - "<h1><center>Welcome to the ITNSA.MY</center></h1>"

## 09- ftp.yml

- Create a playbook called 09-ftp.yml for configuring the ftp service.
  - Install and configure vsftpd in Linux
    - Create user ftpuser with password of Skills39
    - Make sure the users are jailed in their respective home directory
    - Use certificate generate by ADCS

## 10-net-backup.yml

- Create a playbook called 10-net-backup.yml to save and backup configurations of network devices.
  - Create TFTP server **manually** at EVE-NG/Ansible server (tftpd-hpa)
    - Save all configuration at folder /data/ansible/backup.
    - Make sure the TFTP is allowed to create new files.
  - Make sure every network devices configuration is saved.
    - Save the configuration in the folder with the format "\$h\$t.cfg"

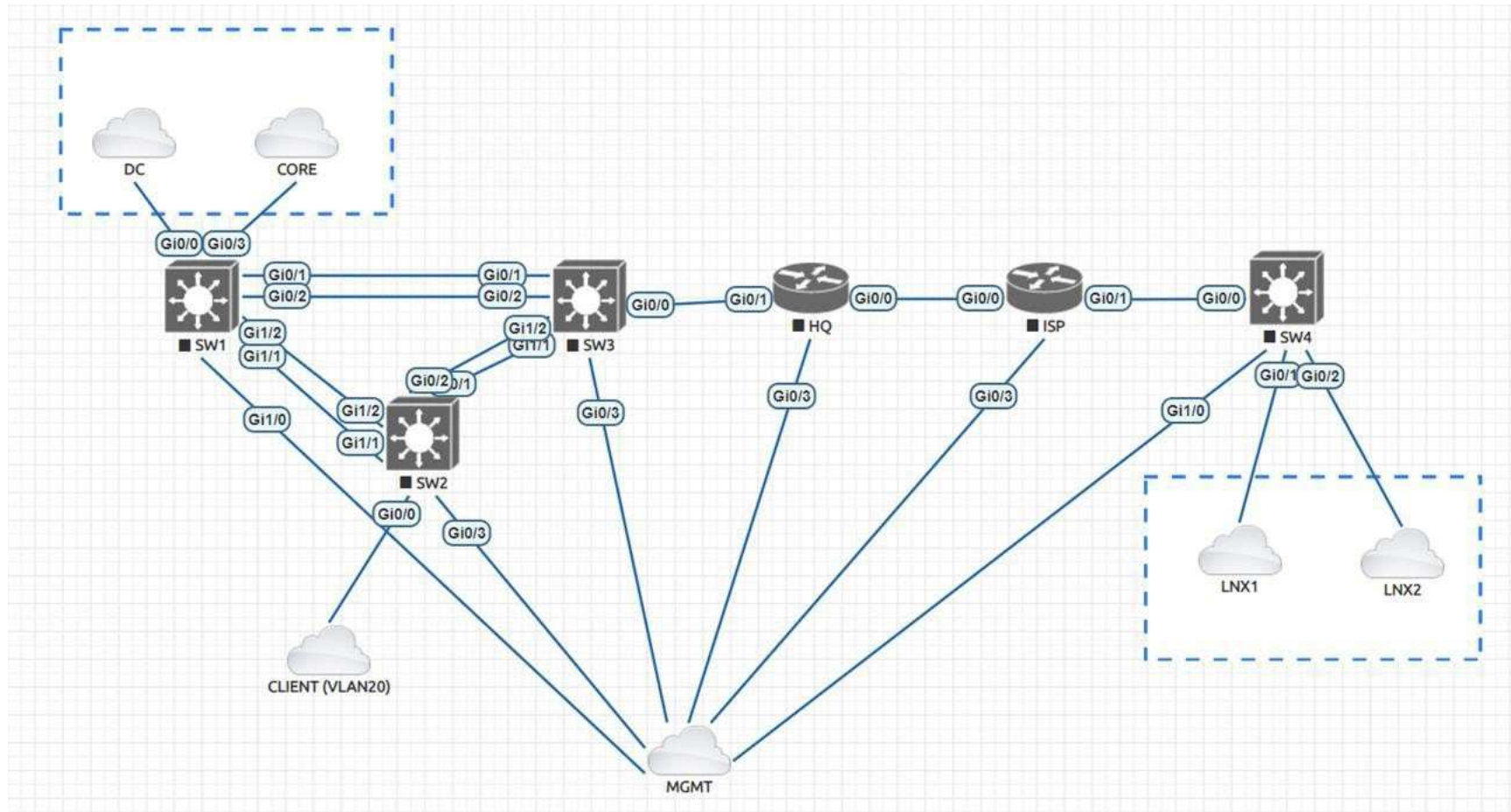
## NETWORK ADDRESS TABLE

| Device | Operating System           | Interface              | Address                   |
|--------|----------------------------|------------------------|---------------------------|
| DC     | Windows Server 2019 (GUI)  | Ethernet 1             | 172.16.10.10 /24          |
| CORE   | Windows Server 2019 (Core) | Ethernet 1             | 172.16.10.11 /24          |
| LNK1   | Debian 12.5 CLI            | Ethernet 0             | 172.16.20.10 /24          |
| LNK2   | Debian 12.5 CLI            | Ethernet 0             | 172.16.20.11 /24          |
| CLIENT | Windows 10 Enterprise      | Ethernet 0             | 192.168.10.XXX /24 (DHCP) |
| HQ     | Cisco vIOS Router          | GigabitEthernet 0/1.10 | 172.16.10.254 /24 (SRV)   |
|        |                            | GigabitEthernet 0/1.20 | 192.168.10.254 /24 (LAN)  |
|        |                            | GigabitEthernet 0/1.99 | 172.16.99.6 /29 (MGMT)    |
|        |                            | GigabitEthernet 0/0    | 24.48.72.1 /28            |
|        |                            | Loopback 100           | 1.1.1.1 /32               |
| ISP    | Cisco vIOS Router          | GigabitEthernet 0/0    | 24.48.72.14 /28           |
|        |                            | GigabitEthernet 0/1    | 172.16.20.254 /24         |
|        |                            | Loopback 100           | 2.2.2.2 /32               |
| SW1    | Cisco vIOS Switch L3       | VLAN 99                | 172.16.99.1 /29           |
| SW2    | Cisco vIOS Switch L3       | VLAN 99                | 172.16.99.2 /29           |
| SW3    | Cisco vIOS Switch L3       | VLAN 99                | 172.16.99.3 /29           |

## MANAGEMENT NETWORK ADDRESS TABLE

| Device | Address       |  | Device | Address       |
|--------|---------------|--|--------|---------------|
| DC     | 10.1.1.11 /24 |  | SW2    | 10.1.1.32 /24 |
| CORE   | 10.1.1.12 /24 |  | SW3    | 10.1.1.33 /24 |
| LNK1   | 10.1.1.21 /24 |  | SW4    | 10.1.1.34 /24 |
| LNK2   | 10.1.1.22 /24 |  | HQ     | 10.1.1.41 /24 |
| SW1    | 10.1.1.31 /24 |  | ISP    | 10.1.1.42 /24 |

## TOPOLOGY







## PRE-CONFIG

/etc/ansible/ansible.cfg

[defaults]

inventory=/etc/ansible/hosts  
remote\_user=root host\_key\_checking=False  
become=True become\_user=root  
become\_ask\_pass=False  
ansible\_python\_interpreter=/usr/bin/python3

=====

/etc/ansible/hosts

all: hosts: #Linux Hosts LNX1: ansible\_host: 10.1.1.21 hostname:  
"LNX1" network\_addr: "172.16.20.10" webmessage:  
"<h1><center>Welcome to the world of automation with ansible" webcolor:  
"red" LNX2: ansible\_host: 10.1.1.22 hostname: "LNX2"  
network\_addr: "172.16.20.11" webmessage: "<h1><center>How exciting to  
deploy services using automation" webcolor: "green" #Windows Hosts  
WIN1: ansible\_host: 10.1.1.11 hostname: "DC" network\_addr:  
"172.16.10.10" WIN2: ansible\_host: 10.1.1.12 hostname: "CORE"  
network\_addr: "172.16.10.11" #Cisco Switch Hosts RTR1: ansible\_host:  
10.1.1.41 hostname: "HQ" RTR2: ansible\_host: 10.1.1.42 hostname:  
"ISP" SW1: ansible\_host: 10.1.1.31 hostname: "SW1" SW2:  
ansible\_host: 10.1.1.32 hostname: "SW2" SW3: ansible\_host: 10.1.1.33  
hostname: "SW3" SW4: ansible\_host: 10.1.1.34 hostname: "SW4"  
children: linux: hosts:  
LIN1:  
LIN2:  
windows:  
hosts:  
WIN1:  
WIN2:  
cisco:  
children:  
routers:  
hosts:  
RTR1:

RTR2:

switches:

hosts:

SW1:

SW2:

SW3:

SW4:

"/etc/ansible/group\_vars/cisco"

ansible\_user: itnsa ansible\_ssh\_pass:  
Skills39 ansible\_password: Skills39  
ansible\_become: yes  
ansible\_become\_method: enable  
ansible\_connection: network\_cli  
ansible\_network\_os: cisco.ios.ios  
ansible\_network\_cli\_ssh\_type: libssh

=====

"/etc/ansible/group\_vars/linux"

```
ansible_user: root ansible_password:
Skills39 ansible_become: true
ansible_become_method: su
ansible_become_password: Skills39
```

```
=====
"/etc/ansible/group_vars/windows"
```

```
ansible_user: Administrator ansible_password:
Skills39 ansible_connection: winrm
ansible_winrm_server_cert_validation: ignore
ansible_ssh_port: 5986
```

```
=====
CISCO PRE-CONFIG
```

```
hostname DSW1 ip domain name example.net username itnsa
privilege 15 secret Skills39 crypto key generate rsa general-keys
modulus 2048 label BOOTKEYS ip ssh version 2
line vty 0 530
    transport input ssh
login local
ip http secure-server ip http
authentication local interface
GigabitEthernet 1/0 no sw ip
address 10.1.1.31 255.255.255.0
no shutdown
exit
```

```
hostname DSW2 ip domain name example.net username itnsa
privilege 15 secret Skills39 crypto key generate rsa general-keys
modulus 2048 label BOOTKEYS ip ssh version 2
line vty 0 530
    transport input ssh
login local
ip http secure-server ip http
authentication local interface
GigabitEthernet 0/3 no sw ip
address 10.1.1.32 255.255.255.0
no shutdown
exit
```

```
hostname DSW3 ip domain name example.net username itnsa
privilege 15 secret Skills39 crypto key generate rsa general-keys
modulus 2048 label BOOTKEYS ip ssh version 2
line vty 0 530
    transport input ssh
login local
ip http secure-server ip http
authentication local interface
GigabitEthernet 0/3 no sw ip
address 10.1.1.33 255.255.255.0
no shutdown
exit
```

```
hostname DSW4 ip domain name example.net username itnsa
privilege 15 secret Skills39 crypto key generate rsa general-keys
modulus 2048 label BOOTKEYS ip ssh version 2
line vty 0 530
    transport input ssh
login local
```

```
ip http secure-server ip http
authentication local interface
GigabitEthernet 1/0 no sw ip
address 10.1.1.34 255.255.255.0
no shutdown
exit
```

```
hostname DRTR1 ip domain name example.net username itnsa
privilege 15 secret Skills39 crypto key generate rsa general-keys
modulus 2048 label BOOTKEYS ip ssh version 2
line vty 0 530
    transport input ssh
login local
ip http secure-server ip http
authentication local interface
GigabitEthernet 0/3 ip address
10.1.1.41 255.255.255.0
no shutdown
exit
```

```
hostname DRTR2 ip domain name example.net username itnsa
privilege 15 secret Skills39 crypto key generate rsa general-keys
modulus 2048 label BOOTKEYS ip ssh version 2
line vty 0 530
    transport input ssh
login local
ip http secure-server ip http
authentication local interface
GigabitEthernet 0/3 ip address
10.1.1.42 255.255.255.0
no shutdown
exit
```